

Module 4 Lab 1: Describe and Correlate

CS 82A - Introduction to Data Science

Ramisha Tasfia

GitHub: https://github.com/ramishatasfia5156-commits/cs82a-coursework/blob/main/module4/module4_lab.ipynb

Load the Clean Data

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt
import statistics

df = pd.read_csv('sales_clean.csv')
print('Shape as loaded:', df.shape)
```

Shape as loaded: (292, 6)

```
In [2]: # The sales_clean.csv in my repo was pushed before the Module 3 Lab 1 step
        # that removed
        # the 3 negative-quantity orders (1140, 1233, 1025). Applying that same
        # documented step here.
        print(df[df['qty'] < 0])
        df = df[df['qty'] >= 0]
        df['zip'] = df['zip'].astype(str).str.zfill(5) # zip codes are
        # identifiers, not numbers
        print()
        print('Shape after removing negative qty:', df.shape)
```

	order_id	date	product	price	qty	zip
199	1140	2026-04-18	webcam	38.69	-5	98101
255	1233	2026-05-09	desk lamp	22.64	-4	10001
289	1025	2026-04-27	keyboard	47.30	-2	2116

Shape after removing negative qty: (289, 6)

Center and Spread of Price and Qty

```
In [3]: summary = pd.DataFrame({
        'mean': [df['price'].mean(), df['qty'].mean()],
        'median': [df['price'].median(), df['qty'].median()],
        'std': [df['price'].std(), df['qty'].std()],
    }, index=['price', 'qty']).round(2)
summary
```

```
Out[3]:
```

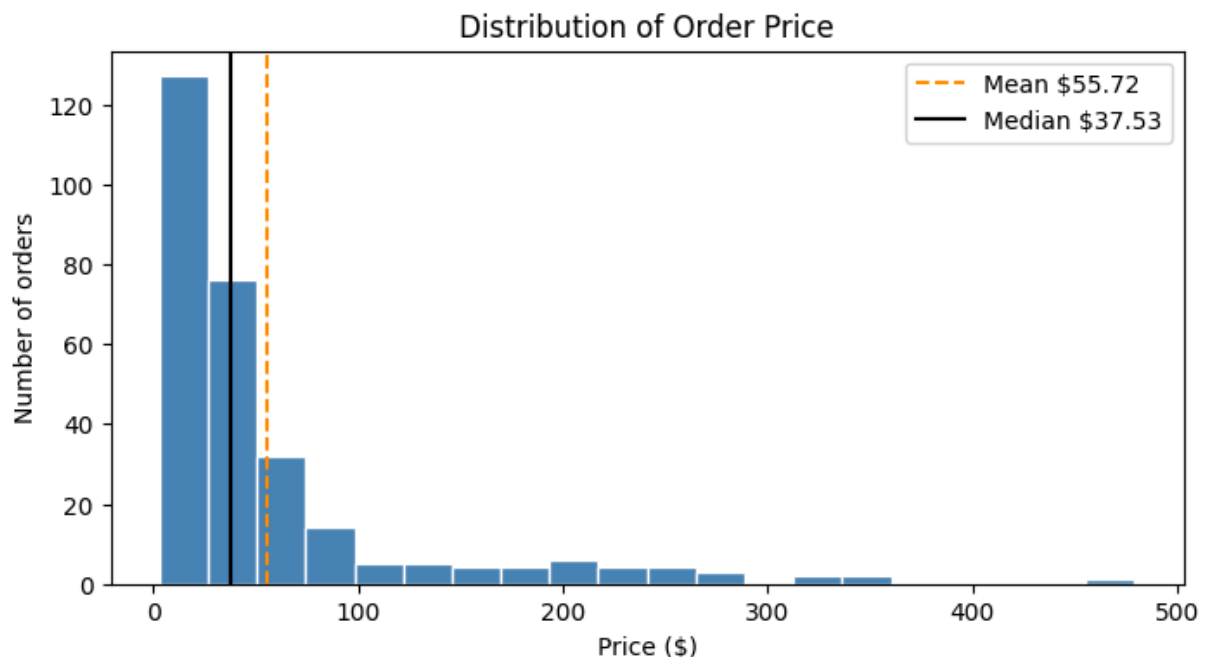
	mean	median	std
price	55.72	37.53	72.11
qty	2.98	3.00	1.41

Which Summary Would I Report?

For price I would report the median (\$37.53), instead of the mean (\$55.72). The mean is about \$18 higher than the median because of a small number of very expensive monitors (up to \$479), which are pulling it upward. Therefore, the mean describes a price that most individual orders will not reach. The median better represents the price of a typical order in this data set because it is not affected by the high-priced values at the tail.

Histogram of Price

```
In [4]: fig, ax = plt.subplots(figsize=(8, 4))
ax.hist(df['price'], bins=20, color='steelblue', edgecolor='white')
ax.axvline(df['price'].mean(), color='darkorange', linestyle='--',
label=f"Mean ${df['price'].mean():.2f}")
ax.axvline(df['price'].median(), color='black', linestyle='-',
label=f"Median ${df['price'].median():.2f}")
ax.set_title('Distribution of Order Price')
ax.set_xlabel('Price ($)')
ax.set_ylabel('Number of orders')
ax.legend()
plt.show()
```



Shape: right-skewed. Most prices are under \$100, with a long tail of expensive items to the right. This agrees with reporting the median.

Revenue Column

```
In [5]: df['revenue'] = df['price'] * df['qty']
df.head()
```

```
Out[5]:
```

	order_id	date	product	price	qty	zip	revenue
0	1254	2026-04-06	mug	11.36	4	60614	45.44
1	1057	2026-05-30	charger	14.79	3	30303	44.37
2	1150	2026-05-06	mug	5.50	1	10001	5.50
3	1066	2026-05-30	notebook	37.53	1	98101	37.53
4	1114	2026-04-06	webcam	45.59	4	90405	182.36

Correlation Matrix

```
In [6]: # order_id is a label and zip is now text, so only the measured columns are
        # compared
        corr = df[['price', 'qty', 'revenue']].corr(numeric_only=True).round(3)
        corr
```

```
Out[6]:
```

	price	qty	revenue
price	1.000	-0.040	0.821
qty	-0.040	1.000	0.314
revenue	0.821	0.314	1.000

Strongest off-diagonal value: price vs. revenue, $r = 0.821$ (strong, positive). Qty vs. revenue is $r = 0.314$ (weak, positive). Price vs. qty is about 0, so customers do not buy fewer units of expensive items.

Scatter Plots

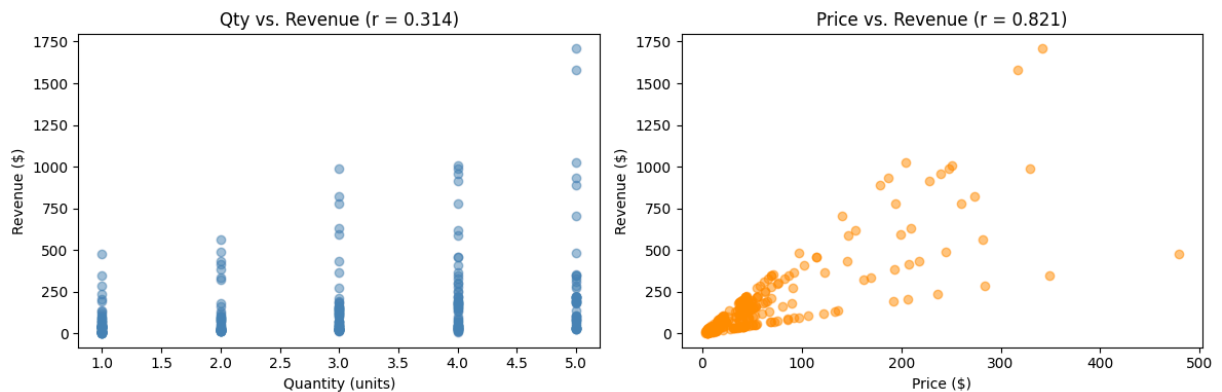
```
In [7]: fig, axes = plt.subplots(1, 2, figsize=(12, 4))

axes[0].scatter(df['qty'], df['revenue'], alpha=0.5, color='steelblue')
axes[0].set_title(f"Qty vs. Revenue (r = {corr.loc['qty', 'revenue']})")
axes[0].set_xlabel('Quantity (units)')
axes[0].set_ylabel('Revenue ($)')

axes[1].scatter(df['price'], df['revenue'], alpha=0.5, color='darkorange')
axes[1].set_title(f"Price vs. Revenue (r = {corr.loc['price', 'revenue']})")
```

```
'revenue']})")
axes[1].set_xlabel('Price ($)')
axes[1].set_ylabel('Revenue ($)')

plt.tight_layout()
plt.show()
```



Both clouds agree with their r values. Qty vs. revenue drifts upward but with a very wide vertical spread at every quantity, which fits a weak r of 0.31. Price vs. revenue rises clearly, fanning out as price grows because quantity varies, which fits a strong r of 0.82.

The Analyst's Sentence

Revenue is largely driven by an order's price rather than the number of units purchased. There is a strong correlation between price and revenue ($r = 0.82$), while there is only a weak correlation between quantity and revenue ($r = 0.31$). However, since revenue is calculated using price, part of this relationship is built into the calculation. Additionally, a small number of expensive monitor orders drive much of the relationship. Therefore, this shows which factor has a stronger relationship with revenue in this particular data set, but it does not mean that increasing prices would necessarily increase revenue.

By-Hand Check

Data: [2, 4, 4, 6, 9]

Mean: $(2 + 4 + 4 + 6 + 9) / 5 = 25 / 5 = 5.0$

Median: the values are already in order and there are 5 of them, so the median is the 3rd value: **4**

```
In [8]: values = [2, 4, 4, 6, 9]
print('Mean: ', sum(values) / len(values))
print('Median:', statistics.median(values))
```

Mean: 5.0

Median: 4

AI Disclosure

AI Disclosure: Claude (Anthropic) was used to help explain concepts and draft initial code for this lab. All code was reviewed, retyped, and understood by me. I can explain why the median is the better summary for right-skewed prices, how the histogram shows that skew, how to read the correlation matrix, and why the correlation between price and revenue needs a caution attached.

Key prompt: "Help me with Module 4 Lab 1: Describe and Correlate using my sales_clean.csv from Module 3."